

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT202		INTELLIGENT SCIENCE	

2nd SEMESTER 2022/23 FINAL EXAMINATION

UNDERGRADUATE – YEAR 3

COMPLEXITY OF ALGORITHMS

TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

- 1. This is a closed-book examination.**
- 2. The examination will be conducted in all other respects in accordance with the Regulations for the Conduct of Examinations.**
- 3. Total marks available are 100. This exam will count for 80% of the final mark.**
- 4. Answer all questions. There is NO penalty for providing a wrong answer.**
- 5. Only English solutions are accepted.**

Question 1. [5 MARKS]

Suppose that we have numbers between 1 and 1000 in a binary search tree, and we want to search for the number 363. Which of the following sequences could not be the sequence of nodes examined? Please explain in detail.

- a. 2, 252, 401, 398, 330, 344, 397, 363.
- b. 924, 220, 911, 244, 898, 258, 362, 363.
- c. 925, 202, 911, 240, 912, 245, 363.

[5 marks]

Question 2. [5 MARKS]

Illustrate the result of each operation in the sequence: PUSH(S, 4), PUSH(S, 1), PUSH(S, 3), POP(S), and PUSH(S, 8) on an initially empty stack S.

[5 marks]

Question 3. [10 MARKS]

Show that the running time of QUICKSORT is

$$\Theta(n^2)$$

when the array contains distinct elements and is already sorted in decreasing order.

[10 marks]

Question 4. [10 MARKS]

Show the process of removal of top element from the heap A[15, 13, 9, 5, 12, 8, 7, 4, 0, 6, 2, 1]. Please show original heap, all intermediate steps and final heap.

[10 marks]

Question 5. [10 MARKS]

Use the master method to prove that the recurrence:

$$T(n) = T(n/2) + \Theta(1)$$

has the solution:

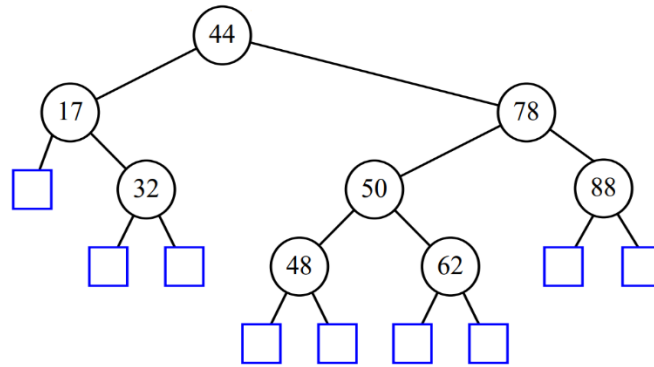
$$T(n) = \Theta(\lg n)$$

[10 marks]

Question 6. [10 MARKS]

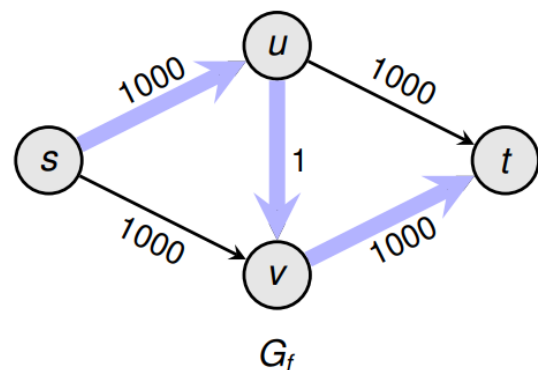
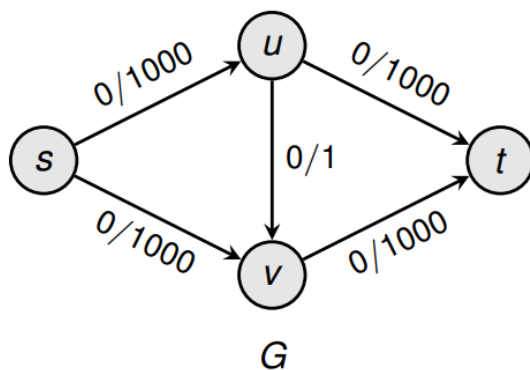
Draw the AVL tree resulting from the insertion of an item with a key 54 into the following AVL tree. Please also show all intermediate steps.

[10 marks]



Question 7. [5+3+7=15 MARKS]

For a given flow network, the Ford-Fulkerson algorithm enables us to find the maximum flow over the network. We have the original flow network on the left where s is the source node, and t is the target node. On the right, we have a highlighted initial path ($s \rightarrow u \rightarrow v \rightarrow t$).



- (1) What's the maximum number of augmenting paths (including the initial path) that need to be found until the termination of the Ford-Fulkerson algorithm given the highlighted initial path ($s \rightarrow u \rightarrow v \rightarrow t$). [5 marks]

Breadth first search from the source to the target node can be used as a deterministic method to choose the augmenting paths in the Ford-Fulkerson algorithm.

- (2) Explain how BFS works in our example, and how many augmenting paths are found? [3 marks]
- (3) Using BFS, we know that the shortest paths length from source node to any other node in the residual network monotonically (non-strictly) increase after each flow update. What is the time complexity of the Ford-Fulkerson for a network with V nodes and E edges if BFS is used? [Hint: how many times can one edge becomes the bottleneck in the residual network?] [7 marks]

Question 8. [7+8+5=20 MARKS]

Alice and Bob wish to use the RSA cryptosystem for communication. Alice announced the public modulus and encryption key pair to be (143, 11).

- (1) Suppose that Bob wish to send a message 2, what is the encrypted message?
[7 marks]
- (2) Derive the private key.
[8 marks]
- (3) Assume Bob sent an encrypted message 4, what's the original unencrypted message?
[5 marks]

Question 9. [5+5+5=15 MARKS]

A Hamiltonian cycle of an undirected graph $G = (V, E)$ is a simple cycle that contains each vertex in V . The Hamiltonian cycle problem asks "Does the undirected graph G have a Hamiltonian cycle?"

The travelling salesman optimization problem asks "Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?"

- (1) States a decision version of the travelling salesman problem.
[5 marks]
- (2) Provide a reduction from the Hamiltonian cycle problem to the travelling salesman decision problem, and justify it.
[5 marks]
- (3) It is known that the Hamiltonian cycle problem belongs to the NP-complete problems. Prove or disprove that the travelling salesman decision problem is NP-complete.
[5 marks]

END OF EXAM PAPER